

NATURAL LANGUAGE PROCESSING

WEEK 1: A HISTORICAL JOURNEY FROM RULES TO STATISTICS TO NEURAL NETWORKS

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WHAT IS NLP?

Teaching computers to understand human language

1 The Challenge

Language is ambiguous, context-dependent, and constantly evolving, making it one of AI's hardest problems.

2 Real-World Impact

Powers search engines, machine translation, virtual assistants, sentiment analysis, and information extraction systems.

3 The Evolution

Journey from hand-crafted rules to statistical learning to automatic feature discovery through neural networks.

AMBIGUITY & CONTEXT

I saw the man with the telescope

- *Did I use a telescope to see the man?*
- *Did I see a man who had a telescope?*

NLP EVOLUTION TIMELINE

Teaching computers to understand human language

1950s-1980s

Rule-based and symbolic approaches with hand-crafted grammars

1980s-2000s

Statistical revolution - learning patterns from data

2000s-2012

Advanced statistical methods with early neural networks

2013-2017

Neural word embeddings, RNNs, and attention mechanisms

2018-present

Transformer era and large language models

EARLY NLP SYSTEMS

Pattern matching and formal grammars

1 ELIZA (1966)

First chatbot using pattern matching and template responses. Demonstrated humans would attribute intelligence to simple systems.

2 SHRDLU (1971)

Natural language understanding in a blocks world. Combined parsing, reasoning, and dialogue in restricted domains.

3 Critical Limitations

Brittle, domain-specific, no learning capability. Required massive manual knowledge engineering that didn't scale.

Pattern: "I am + (sad|depressed|unhappy)"

Response: "I am sorry to hear you are \$1"

Input: "I am feeling sad"

Output: "I am sorry to hear you are sad"

RULE-BASED PARSING

Grammars for analyzing sentence structure

1 Context-Free Grammars

Production rules define syntactic structure. Chart parsing algorithms (CYK, Earley) find all valid parse trees efficiently.

2 Augmented Systems

ATNs and DCGs added conditions and logic programming to handle linguistic complexity beyond simple context-free rules.

3 The Trade-off

Broader coverage meant exponential ambiguity. Every sentence had dozens of grammatically valid but semantically meaningless parses.

Grammar Rules:

$S \rightarrow NP VP$

$NP \rightarrow Det N \mid Det N PP \mid Pronoun$

$VP \rightarrow V NP \mid V NP PP$

$PP \rightarrow P NP$

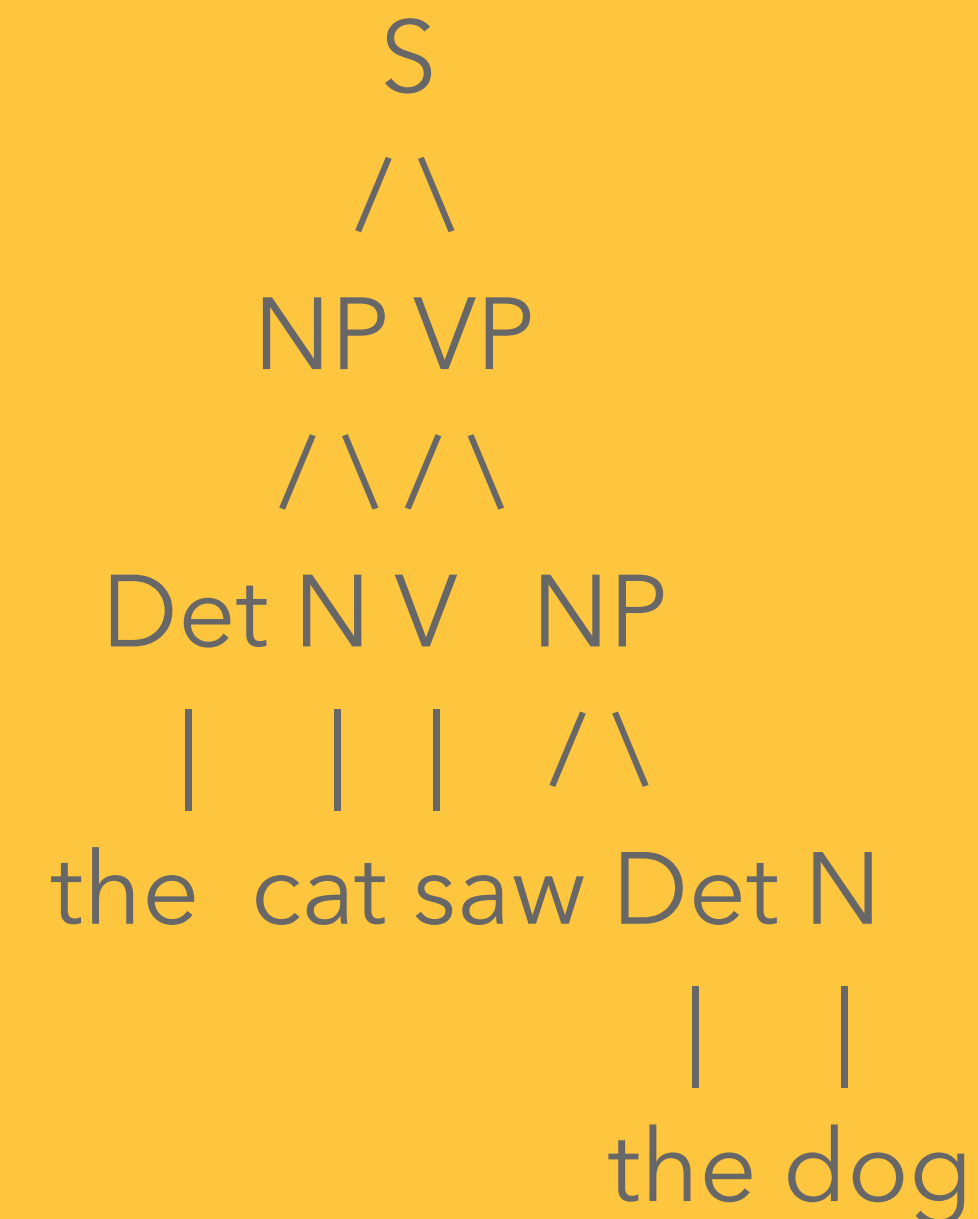
$Det \rightarrow "the" \mid "a"$

$N \rightarrow "cat" \mid "dog" \mid "telescope"$

$V \rightarrow "saw" \mid "ate"$

$P \rightarrow "with" \mid "on"$

Example: "The cat saw the dog"



KNOWLEDGE REPRESENTATION

Semantic relationships in structured form

1 WordNet (1995)

Lexical database organizing words into synonym sets with semantic relations (hypernyms, hyponyms). Still widely used today.

2 FrameNet & PropBank

Frame semantics and semantic role labeling. Captured predicate-argument structures and how words evoke conceptual frames.

3 Applications

Enabled word sense disambiguation, semantic similarity computation, and question answering through structured world knowledge.

Synset: {bank₁, banking_comp, depository_fin_inst}

Hypernym: financial_institution

Hyponym: commercial_bank, savings_bank

Synset: {bank₂, bank_building}

Hypernym: building, edifice

Meronym: vault

Path Similarity:

$\text{sim}(\text{cat}, \text{dog}) = 0.857$

$\text{sim}(\text{cat}, \text{car}) = 0.143$

WHY RULE-BASED SYSTEMS FAILED TO SCALE

1 Knowledge acquisition bottleneck

Required expert linguists to hand-craft every rule

2 Coverage problem

Impossible to enumerate all language phenomena and edge cases

3 Brittleness

Small variations in input completely broke the system

4 Ambiguity explosion

Natural language generates massive numbers of valid parses

5 Domain adaptation

Rules crafted for one domain don't transfer to others

STATISTICAL REVOLUTION

Learning language patterns from data, not rules

1 Shannon's Insight (1948)

Applied information theory to language. Showed language has statistical regularities that can be learned and exploited.

2 N-gram Language Models

Predict next word based on previous n-1 words. Simple but powerful approach using probability and smoothing techniques.

3 Paradigm Shift

From hand-crafted rules to data-driven learning. Let the data tell us about language structure and patterns.

Language Model Chain Rule:

$$P(w_1, w_2, \dots, w_n) = \prod_i P(w_i | w_1, \dots, w_{i-1})$$

Bigram Approximation:

$$P(w_i | w_1, \dots, w_{i-1}) \approx P(w_i | w_{i-1})$$

$$P(w_i | w_{i-1}) = \text{Count}(w_{i-1}, w_i) / \text{Count}(w_{i-1})$$

"I love NLP. I love machine learning. NLP is great."

$$P(\text{love} | I) = 2/2 = 1.0$$

$$P(\text{NLP} | \text{love}) = 1/2 = 0.5$$

$$P(\text{machine} | \text{love}) = 1/2 = 0.5$$

$$\begin{aligned} P(\text{"I love NLP"}) &= P(I) \times P(\text{love}|I) \times P(\text{NLP}|\text{love}) \\ &= 0.2 \times 1.0 \times 0.5 = 0.1 \end{aligned}$$

HIDDEN MARKOV MODELS

Probabilistic model for sequential data

1 Core Insight

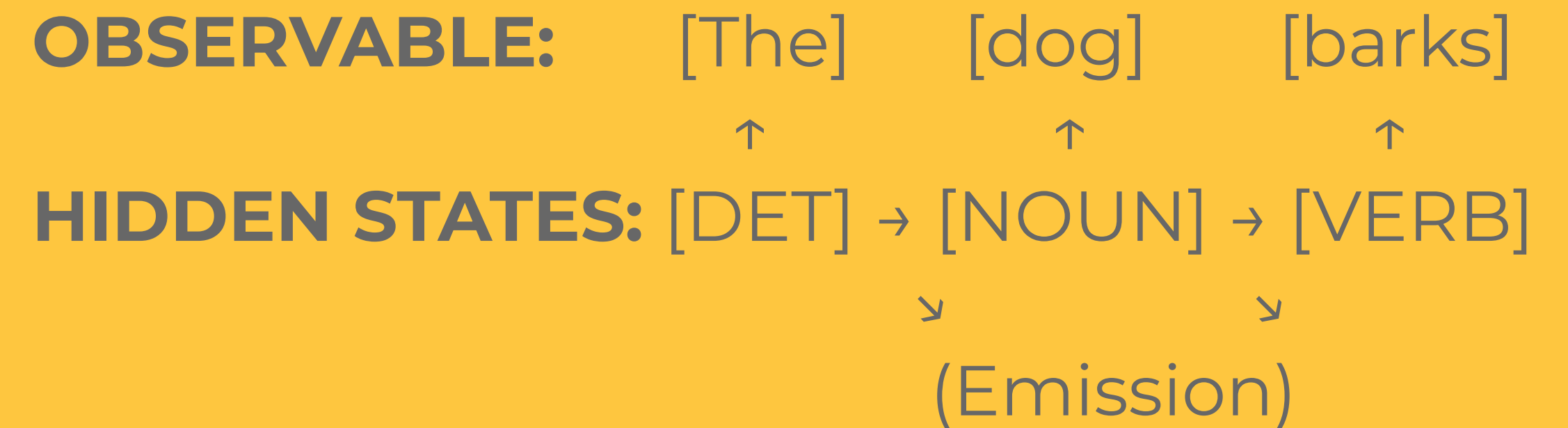
Model sequences where true states are hidden, but they generate observable outputs. Learn patterns from labeled data.

2 Markov Assumption

Current state depends only on previous state, not entire history. Makes computation tractable while capturing dependencies.

3 Importance

First successful probabilistic model for sequence labeling. Unified framework for inference and learning. Dominated NLP 1990s-2000s.



We observe words, but want to infer hidden POS tags

HMMs model both $P(\text{tag sequence})$ and $P(\text{words} \mid \text{tags})$

HMM EXAMPLE

Sentence: "The dog barks"

$O = [\text{The}, \text{dog}, \text{barks}]$

One possible tag sequence: $Q = [\text{DET}, \text{NOUN}, \text{VERB}]$

$$\begin{aligned} P(O, Q \mid \lambda) &= \pi(\text{DET}) \times b_{\text{DET}}(\text{The}) && [\text{initial} \times \text{emit}] \\ &\times a(\text{DET} \rightarrow \text{NOUN}) \times b_{\text{NOUN}}(\text{dog}) && [\text{trans} \times \text{emit}] \\ &\times a(\text{NOUN} \rightarrow \text{VERB}) \times b_{\text{VERB}}(\text{barks}) && [\text{trans} \times \text{emit}] \\ \\ &= 0.3 \times 0.6 && = 0.18 \\ &\times 0.7 \times 0.3 && = 0.21 \times 0.18 = 0.0378 \\ &\times 0.5 \times 0.4 && = 0.20 \times 0.0378 = 0.00756 \end{aligned}$$

$$P(O, Q) = 0.00756$$

! Many tag sequences exist!

Search algorithms are used to find the best one

HMM FORMAL DEFINITION

States: $S = \{s_1, s_2, \dots, s_n\}$ (e.g., {DET, NOUN, VERB, ...}), Observations: $V = \{v_1, v_2, \dots, v_m\}$ (vocabulary)

INITIAL DISTRIBUTION π :

$$\pi_i = P(q_1 = s_i)$$

Probability of starting in state s_i

Example: $\pi(\text{DET}) = 0.3, \pi(\text{NOUN}) = 0.15, \pi(\text{VERB}) = 0.05$

TRANSITION MATRIX A:

$$a_{ij} = P(q_t = s_j \mid q_{t-1} = s_i)$$

Probability of moving from state s_i to s_j

Constraint: $\sum_j a_{ij} = 1$ for all i

Example transition probabilities:

	DET	NOUN	VERB	ADJ
DET	[0	0.7	0.0	0.3]
NOUN	[0	0.1	0.5	0.2]
VERB	[0.1	0.3	0.0	0.1]
ADJ	[0	0.9	0.0	0.0]

EMISSION MATRIX B:

$$b_j(k) = P(o_t = v_k \mid q_t = s_j)$$

Probability of observing word v_k in state s_j

Constraint: $\sum_k b_j(k) = 1$ for all j

Example emissions:

	the	dog	barks	is
DET	[0.6	0.0	0.0	0.3]
NOUN	[0.0	0.3	0.1	0.0]
VERB	[0.0	0.0	0.4	0.5]

HMMs ANSWER 3 QUESTIONS

1 Problem 1: Evaluation

Given model and observations, compute probability $P(O|\lambda)$ — how likely is this sentence.
Use forward algorithm with dynamic programming.

2 Problem 2: Decoding

Given model and observations, find best state sequence of hidden states.
Use Viterbi algorithm.

3 Problem 3: Learning

Given observations, learn parameters λ .
Use Baum-Welch (forward-backward) EM algorithm.

MAXIMUM ENTROPY MODELS

Make fewest assumptions consistent with evidence

1 Core Principle

MaxEnt equals logistic regression.

Discriminative approach that directly models $P(\text{label}|\text{features})$ without independence assumptions.

2 Feature Flexibility

Can incorporate diverse, overlapping features from any source. No need to worry about feature independence.

3 Advantages Over HMMs

Better handles rare events and rich features. Became preferred for classification tasks like POS tagging.

$$P(y | x) = (1/Z(x)) \times \exp(\sum_i \lambda_i f_i(x, y))$$
$$Z(x) = \sum_y \exp(\sum_i \lambda_i f_i(x, y)) \text{ [normalization]}$$

where:

$f_i(x, y)$ = feature functions

λ_i = feature weights (learned)

$Z(x)$ = partition function

Task: Tag word "bank" in "the bank announced"

Features $f(x, y)$:

$f_1: \text{word}=\text{"bank"} \wedge \text{tag}=\text{NOUN} \rightarrow 1$

$f_2: \text{previous_word}=\text{"the"} \wedge \text{tag}=\text{NOUN} \rightarrow 1$

$f_3: \text{next_word}=\text{"announced"} \wedge \text{tag}=\text{NOUN} \rightarrow 1$

$f_4: \text{word}=\text{"bank"} \wedge \text{tag}=\text{VERB} \rightarrow 1$

$f_5: \text{capitalized}=\text{False} \wedge \text{tag}=\text{NOUN} \rightarrow 1$

$$P(\text{NOUN} | \text{"bank"}) \propto \exp(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_5)$$

$$P(\text{VERB} | \text{"bank"}) \propto \exp(\lambda_4)$$

CONDITIONAL RANDOM FIELDS

Gold standard for sequence labeling tasks

1 Motivation

HMMs made strong independence assumptions. CRFs model entire sequence probability discriminatively with global normalization.

2 Key Innovation

Discriminative undirected graphical model. Directly models $P(\text{sequence}|\text{observation})$ rather than joint probability.

3 Dominated Pre-Neural Era

Became state-of-the-art for NER, chunking, and sequence labeling before deep learning.

$$P(y | x) = (1/Z(x)) \times \exp(\sum_t \sum_k \lambda_k f_k(y_{t-1}, y_t, x, t))$$
$$Z(x) = \sum_y \exp(\sum_t \sum_k \lambda_k f_k(y_{t-1}, y_t, x, t))$$

where:

y = label sequence $[y_1, \dots, y_t]$

x = observation sequence $[x_1, \dots, x_t]$

f_k = feature functions (can depend on entire x)

λ_k = weights

Sentence: [John, lives, in, New, York]

Tags: [B-PER, O, O, B-LOC, I-LOC]

Feature functions:

$f_1: y_t = \text{B-PER} \wedge \text{capitalized}(x_t)$

$f_2: y_{t-1} = \text{B-LOC} \wedge y_t = \text{I-LOC}$ [state transition]

$f_3: y_t = \text{B-LOC} \wedge \text{"in"} \in x_{\{t-2:t\}}$ [global context]

$f_4: y_t = \text{B-PER} \wedge x_t \in \text{name_gazetteer}$

FUNDAMENTAL NLP TASKS

1 Sequence labeling

Assign labels to each token (POS tagging, NER, Intent Recognition)

2 Syntactic parsing

Determine grammatical structure of sentences

3 Semantic analysis

Extract meaning representations from text

4 Question Answering

Provide direct answers to questions given some amount of documentation

PART-OF-SPEECH TAGGING

Assigning grammatical categories to words

1 The Challenge

Words are ambiguous. 'I can can the can' - same word, different parts of speech. Context determines meaning.

2 Classical Approaches

Statistical methods (HMMs, MaxEnt) achieved ~97% accuracy for English.

3 Foundation Task

POS tags are features for parsing, information extraction, and machine translation. Critical preprocessing step.

Sentence: "The can can hold water"
Tags: DET N MD VB N

Ambiguity:
can1 (noun): container
can2 (modal verb): ability

NAMED ENTITY RECOGNITION

Identifying and classifying named entities in text

1 The Challenge

Extract persons, organizations, locations, dates from text. Entity boundaries aren't obvious and types are context-dependent.

2 BIO Encoding

Beginning, Inside, Outside tags for multi-word entities. Transforms NER into sequence labeling problem.

3 Feature Engineering

CRFs with rich features: capitalization, prefixes/suffixes, gazetteers, POS tags. State-of-the-art pre-deep learning.

```
Apple Inc. CEO Tim Cook announced products in Cupertino|
B-ORG I-ORG O B-PER I-PER O O O B-LOC
```

```
Washington crossed the Delaware
[George Washington]_PER crossed the [Delaware]_LOC
```


CONSTITUENCY PARSING

Discovering hierarchical phrase structure

1 Parse Trees

Nested constituents (NP, VP, PP) form tree structure. Shows how words group into phrases and clauses.

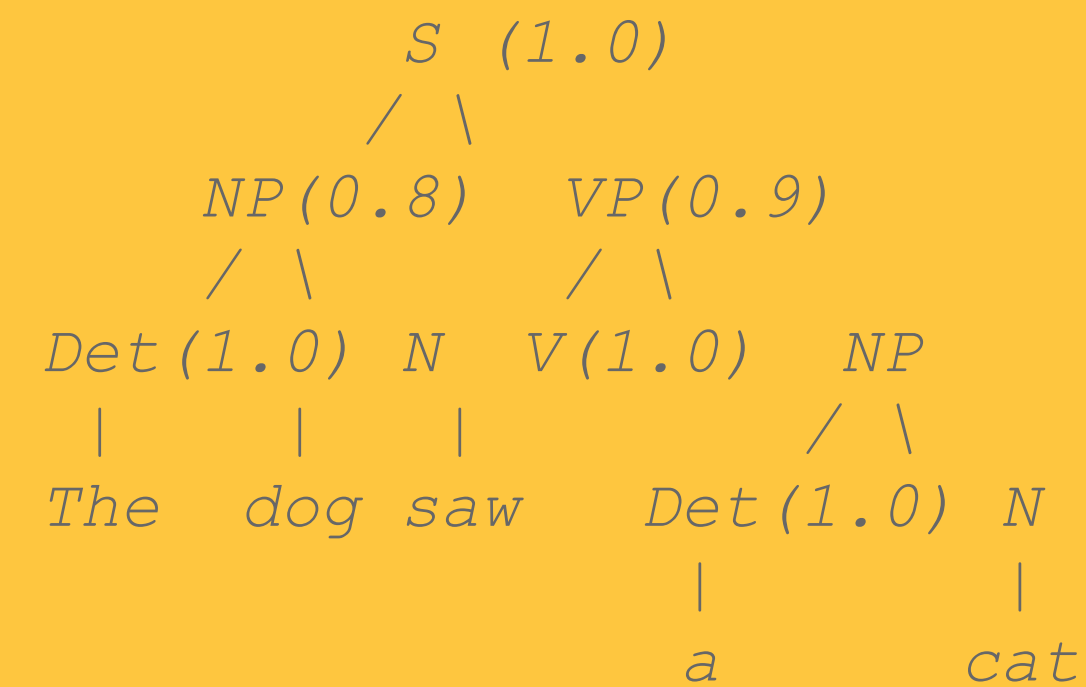
2 Evolution

From rule-based CFGs to probabilistic PCFGs to lexicalized statistical parsers (Collins 1997, Charniak 2000)

3 Berkeley Parser

Achieved state-of-the-art without manual feature engineering.

Sentence: "The dog saw a cat"



$$\begin{aligned} P(\text{tree}) &= P(S \rightarrow NP \ VP) \times P(NP \rightarrow Det \ N) \times P(VP \rightarrow V \ NP) \\ &\times P(NP \rightarrow Det \ N) \times P(Det \rightarrow the) \times P(N \rightarrow dog) \dots \\ &= 0.8 \times 0.6 \times 0.7 \times 0.6 \times 1.0 \times 0.3 \times \dots \end{aligned}$$

DEPENDENCY PARSING

Identifying grammatical relations between words

1 Directed Relations

Graph with words as nodes, grammatical relations as labeled edges (subject, object, modifier).

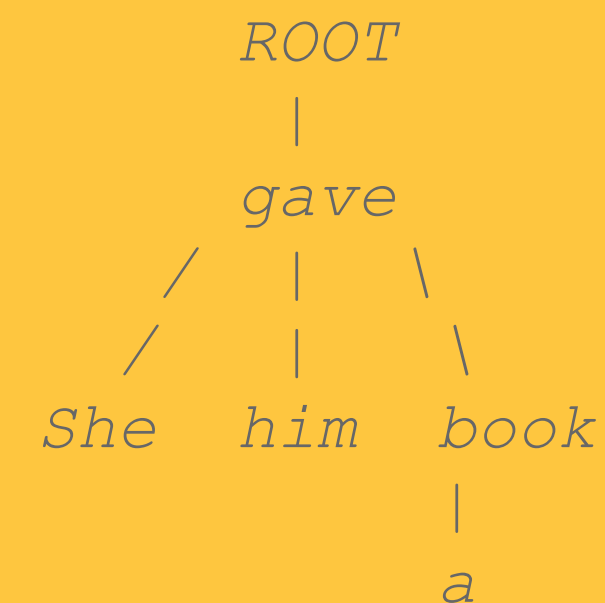
2 Benefit

Understanding deep relationships between words as defined in the sentence

3 Cross-Lingual Standard

Universal Dependencies provides consistent annotation across 100+ languages.

Sentence: "She gave him a book"



Relations:

```
ROOT → gave
gave --nsubj--> She
gave --iobj--> him
gave --obj--> book
book --det--> a
```

SEMANTIC PARSING

Mapping language to formal meaning representations

1 Formal Representations

Convert natural language to logical forms, SQL queries, or executable programs.
Bridges language and computation.

2 Compositionality Challenge

Meaning of whole from meaning of parts.
CCG parsing and statistical methods learn compositional semantics from examples.

3 Applications

Powers question answering over databases, instruction following, and knowledge base population systems.

Natural Language: "Which states border Texas?"

Logical Form (λ -calculus):
 $\lambda x.state(x) \wedge borders(x, Texas)$

SQL:
*SELECT s.name
FROM states s, borders b
WHERE s.id = b.state1
AND b.state2 = 'Texas'*

SPARQL (Knowledge Base):
*SELECT ?state WHERE {
 ?state rdf:type dbo:State .
 ?state dbo:borders dbr:Texas .
}*

QUESTION ANSWERING

Direct interface to information without document retrieval

1 Covers almost all NLP tasks in one

Requires question understanding, information retrieval, NER, semantic parsing, and answer ranking working together

2 QA Types

Factoid (Who invented X?), definitional (What is Y?), and non-factoid questions requiring reasoning across multiple facts.

3 Historical Evolution

Rule-based systems (1960s-80s)

IR-based + shallow NLP (1990s-2000s)

Watson's ensemble (2011)

Neural methods (2013+).

BASEBALL - 1961

First pattern matching to Query system

1 Fixed Patterns

Can't handle complex phrases or paraphrases

2 Manual pattern crafting

On July 7, where was the Red Sox game?" → fails

Q: "Where did the Red Sox play on July 7?"

Pattern matching:

TEAM + "play" + LOCATION + DATE

→ Query database:

SELECT location

FROM games

WHERE team='Red Sox' AND date='July 7'

A: "Fenway Park"

LUNAR - 1972

First semantic approach to QA

1 Specifically built for moon rock questions

Could handle 78% of geologist queries in domain, 0 outside domain

2 Built by Ron Kaplan et al.

QUESTION → WHAT-QUERY | WHERE-QUERY | HOW-MANY-QUERY

WHAT-QUERY → "what is" PROPERTY "of" SAMPLE

PROPERTY → "age" | "composition" | "density"

SAMPLE → "sample" NUMBER | ROCK-TYPE

IBM WATSON - 2011

Ensemble of 100+ algorithms working in parallel

1 DeepQA Architecture

Not one algorithm but massive parallelization. Question analysis, hypothesis generation from multiple sources, 50+ scoring algorithms, ensemble learning.

2 Ensemble Power

No single algorithm works for all question types. Different evidence sources complement each other. Logistic regression learns which scorers to trust.

3 Impact & Scale

2,880 cores, 15TB RAM, 200M pages, 3-second response. Demonstrated real-world viability of statistical NLP at scale.



EVOLUTION OF QA OVER YEARS

1 TREC QA (1999-2007)

Factoid questions over news documents, standardized evaluation

2 WebQuestions (2013)

Questions answerable from Freebase knowledge base

3 WikiQA (2015)

Question-answer pairs derived from Wikipedia articles

4 MS MARCO (2016)

Large-scale dataset with real user search queries

5 Winograd Schema (2011-2018)

An alternative to Turing Test to challenge machine intelligence

An example by Terry Winograd

*"The city council refused the demonstrators a permit because **they** feared violence."*

→ "they" = the city council

*"The city council refused the demonstrators a permit because **they** advocated violence."*

→ "they" = the demonstrator

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5 Winograd Schema

Rules crafted for one domain don't transfer to others

PAPERS TO READ

1 Advances in natural language processing

<https://cs224d.stanford.edu/papers/advances.pdf>

2 Building Watson: An Overview of the DeepQA Project (2013)

<https://cbmm.mit.edu/sites/default/files/documents/watson.pdf>

3 A Statistical Approach to Machine Translation

<https://aclanthology.org/J90-2002.pdf>